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BATCH-	16 B.Tech C.S.E

Programming In C-HDOC DAY_6

Problem_1-

Write a C program to find the radius, diameter, and area of a circle when its circumference is given.

INPUTS

User need to provide the circumference of Circle.

FORMULA USED

As given in the problem ,we need to find radius,diameter and area of circle using circumference.

- Diameter=circumference/pie.
- Radius = diameter/2.
- Area =(pie*radius^2).

OUTPUTS

The program will return Diameter,radius and area of the circle.

ALGORITHM FOR PROBLEM

1. **Start**
2. Declare variables for circumference, radius, diameter, and area.
3. Input the circumference of the circle.
4. Calculate the radius using:

$$radius = \frac{circumference}{2\pi}$$

5. Calculate the diameter using:

$$diameter = 2 \times radius$$

6. Calculate the area using:

$$area = \pi \times radius \times radius$$

7. Print the radius, diameter, and area.
8. **Stop**.

HDOC DAY 6-REPORT

C PROGRAM

```
#include <stdio.h> int
main()
{
   float radius, area, circumference, pi=3.14, diameter;    printf("enter the circumference of the
circle\n");    scanf("%f", &circumference);    diameter=circumference/pi;    radius=diameter/2;
area=pi*radius*radius;    printf("area ,radius and diameter of the circle are %f,%f and%f\n",
area,radius,diameter);    return 0;
}
```

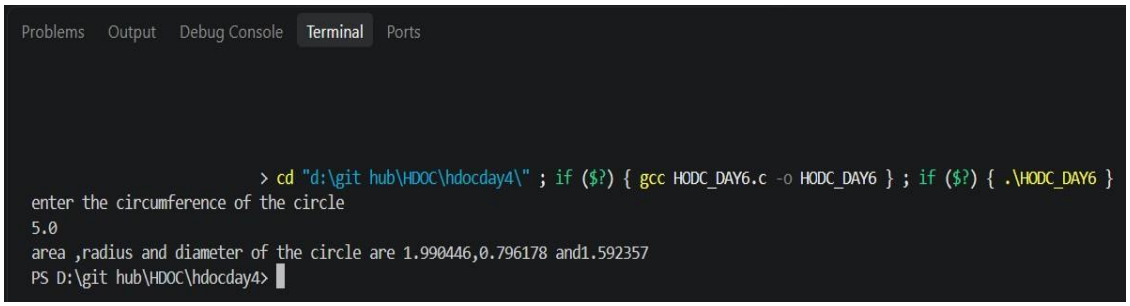
TEST CASES

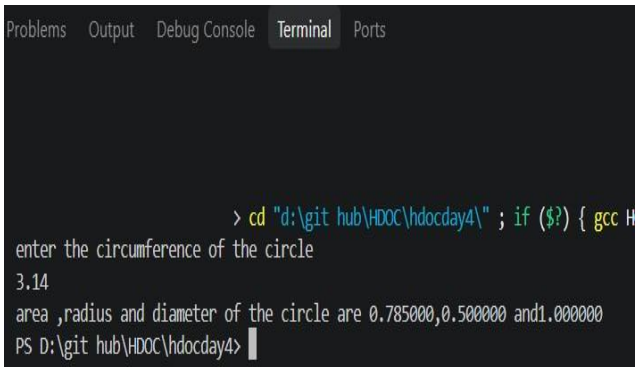
TERMINAL WINDOWS

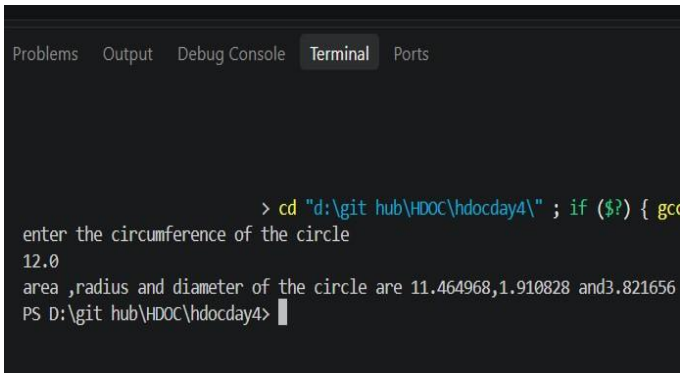
Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
1	3.14	Area ,Radius and Diameter are 0.785000,0.500000 and1.000000.	area ,radius and diameter of the circle are 0.78,0.5 and1.0	PASS
2	12.0	Area ,Radius and Diameter are 11.464968,1.910828 and3.821656	Area ,Radius and Diameter are 11.47,1.910828 and3.821656	PASS

3	5.0	Area ,Radius and Diameter are 1.990446,0.796178 and1.592357	Area ,Radius and Diameter are 1.99,0.79 and1.59	PASS
4	7.0	Area ,Radius and Diameter are 3.901274,1.114650 and2.229299	Area ,Radius and Diameter are 3.9,1.11 and2.22	PASS









Problem_2

Write a C program to find the painting cost and area of an annulus when its outer radius, inner radius, and painting rate per square meter are given.

INPUTS

User needs to provide cost of painting per unit area of annulus, inner and outer radius of annulus.

FORMULAS NEEDED

As mentioned in the problem we need to calculate the Total Painting Cost and Area of the Annulus-

- $Area = \pi(R^2 - r^2)$
- Total Painting Cost=area*painting cost per unit area(in sqm)

OUTPUT

The program will return the area and total painting cost.

ALGORITHM FOR PROGRAM

1. **Start**
2. Declare variables or, ir, rate, area, and cost.
3. Input the **outer radius** of the annulus.
4. Input the **inner radius** of the annulus.
5. Input the **painting rate per square meter**.
6. Calculate the area of the annulus: $Area = \pi(R^2 - r^2)$

where R is the outer radius and r is the inner radius.

7. Calculate the painting cost:

$$Cost = Area \times Rate$$

8. Display the **area of the annulus**.
9. Display the **painting cost**.
10. **Stop**.

PROGRAM

HDOC DAY 6-REPORT

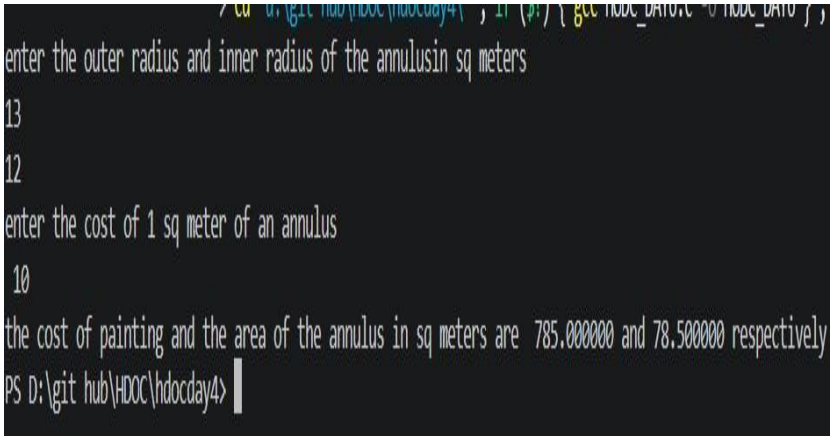
```
#include <stdio.h> int
main()
{
   float or,ir,area,cost,rate;     printf("enter the outer radius and inner radius of the
annulusin sq meters\n");     scanf("%f%f", &or, &ir);     area=3.14*(or*or-ir*ir);
   printf("enter the cost of 1 sq meter of an annulus\n ");     scanf("%f",&rate);
cost=area*rate;     printf("the cost of painting and the area of the annulus in sq meters
are % f and %f respectively",cost,area);
```

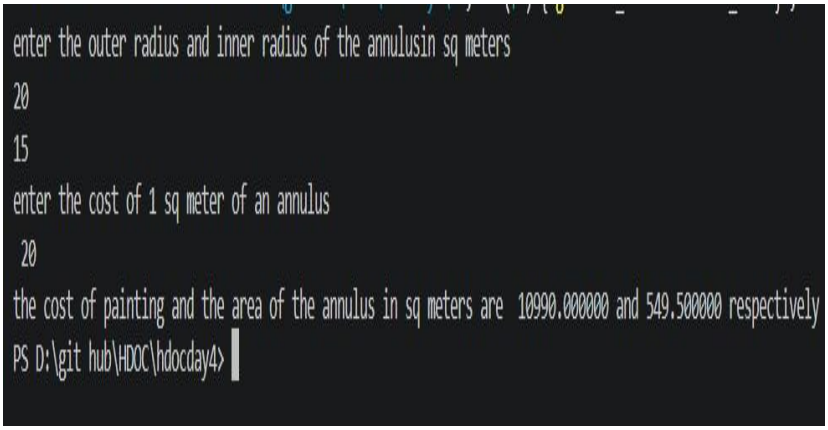
Input	Expected Output	Actual Output	Result (Pass/Fail)
enter the outer radius and inner radius of the annulus in sq meters. 13 12 enter the cost of 1 sq meter of an annulus. 10	the cost of painting and the area of the annulus in sq meters are 785.Oand 78.5 respectively.	the cost of painting and the area of the annulus in sq meters are 785.000000 and 78.500000 respectively.	PASS
enter the outer radius and inner radius of the annulus in sq meters 20 15 enter the cost of 1 sq meter of an annulus 20	the cost of painting and the area of the annulus in sq meters are 10990.0and 549.5respectively.	the cost of painting and the area of the annulus in sq meters are 10990.000000 and 549.500000 respectively.	PASS
enter the outer radius and inner radius of the annulusin sq meters 30 16 enter the cost of 1 sq meter of an annulus 25	the cost of painting and the area of the annulus in sq meters are 50554.0and 2022.16 respectively.	the cost of painting and the area of the annulus in sq meters are 50554.000000 and 2022.160034 respectively.	PASS

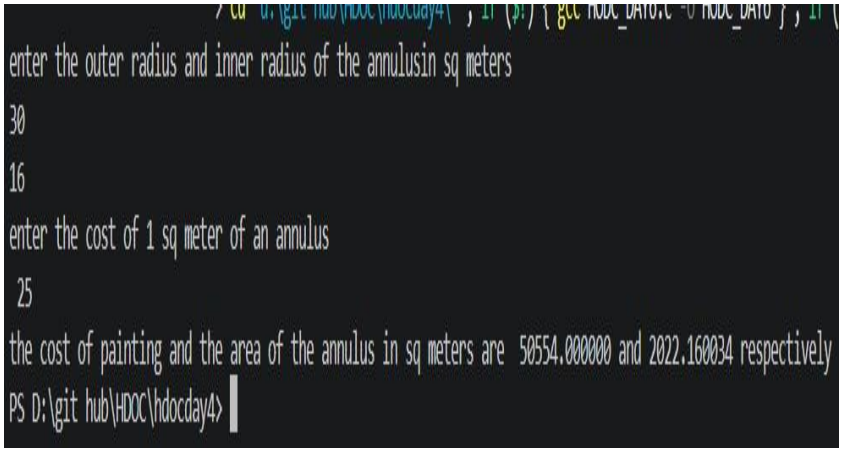
```
}
```

TEST CASES

TERMINAL WINDOWS






enter the outer radius and inner radius of the annulusin sq meters
30
16
enter the cost of 1 sq meter of an annulus
25
the cost of painting and the area of the annulus in sq meters are 50554.000000 and 2022.160034 respectively
PS D:\git hub\HDOC\hdocday4>